

Weekly Research Report: Literature Review on Face Recognition-Based Attendance Systems

Md. Arebi Sarker

Department of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
arebi.sarker@northsouth.edu

Md. Rizvi Ahmed

Department of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
rizvi.safin@northsouth.edu

Safayat Ibrahim

Department of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
safayat.ibrahim@northsouth.edu

Rafiqul Islam

Department of Electrical & Computer Engineering
North South University
Dhaka, Bangladesh
rafiqul.islam12@northsouth.edu

Abstract—This weekly report presents the literature review conducted during the first phase of the CampusEYE project. Six recent studies on facial recognition-based attendance systems were analyzed to compare existing algorithms, identify current research trends, and determine a suitable architecture for developing an accurate, secure, and automated attendance management system.

Index Terms—Face Recognition, Attendance System, ArcFace, InsightFace, Computer Vision

I. INTRODUCTION

The first week of the CampusEYE project focused on reviewing recent research related to facial recognition-based attendance systems. The objective was to understand existing methodologies, compare different recognition techniques, and identify an effective approach for developing a reliable and scalable attendance management system.

II. LITERATURE REVIEW

Paper 1: Automatic Student Attendance System using PCA and Haar Cascade

This study employed Haar Cascade for face detection and PCA (Eigenfaces) for recognition. The approach is lightweight and computationally efficient but its accuracy decreases under varying lighting conditions and facial orientations.

Paper 2: Attendance System using LBPH

The authors combined Haar Cascade with Local Binary Pattern Histogram (LBPH) for real-time face recognition. Although suitable for small-scale applications, recognition performance remains sensitive to environmental variations.

Paper 3: LabVIEW-Based Attendance System

A LabVIEW implementation using pattern matching demonstrated automated attendance recording with a graphical inter-

face. However, the system lacks scalability and robustness for practical deployment.

Paper 4: InsightFace with ArcFace

This work utilized InsightFace and ArcFace to generate facial embeddings for cosine similarity matching. The proposed approach achieved higher recognition accuracy and greater robustness against pose and illumination changes.

Paper 5: CNN vs. LBPH – Systematic Literature Review

The review compared traditional LBPH with CNN-based approaches and concluded that deep learning models consistently outperform handcrafted feature extraction methods under real-world conditions.

Paper 6: Hybrid RFID and Face Recognition

The proposed framework integrated RFID authentication with facial recognition to prevent proxy attendance. Dual-factor verification significantly improved attendance security and reliability.

III. KEY FINDINGS

The reviewed studies show that deep learning-based methods outperform traditional facial recognition techniques in terms of accuracy, robustness, and scalability. Among the reviewed approaches, InsightFace with ArcFace offers the most reliable solution for real-time attendance management, while hybrid RFID integration further improves system security.

IV. CONCLUSION

The first week's literature review provided a solid foundation for the CampusEYE project. Based on the findings, InsightFace with ArcFace has been selected as the most suitable approach for developing an accurate, secure, and scalable automated attendance system.